Ready to Train? — design summary

An R Shiny look-back at how sleep, training and everyday habits line up with feeling ready to train · GEOM90007 Assignment 2

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PMData · 16 participants · 1 Nov 2019 – 1 Apr 2020
3 pages · 5 summary cards · 18 habits · 13 tracked signals

1 Purpose and audience

This dashboard comes from a system I have been building: a ***personal digital twin*** with two linked parts, which I call the ***cognitive core*** (my knowledge, goals, preferences and long-term judgements) and the ***data core*** (my measured body and daily life: sleep, training, steps, heart rate and how I feel). Together they would let an AI understand both what I value and how I am right now.

My earlier prototype collected data well but gave little thought to visual hierarchy, data integrity or user tasks, so this assignment redesigns the data core around three questions: **How is the body today? Which records is that judgement based on? Which habits go with a better next day?** It is an exploratory look-back for people who log training, sleep and how they feel (recreational athletes, regular gym-goers) and their coaches — not medical or training advice.

2 Data and its limits

My own records are not yet long or complete enough, so the prototype uses **PMData**, which has the same structure: 16 active adults with Fitbit data, morning check-ins, training logs and food questionnaires. **NHANES** adds outside context on sleep length only. In my long-term system people would summarise their day in the evening; PMData check-ins were made in the morning, and the dashboard keeps that timing instead of claiming the future design.

3 Integrity and findings

Associations remove each participant's own average, use participant-clustered 95% intervals and a Benjamini–Hochberg correction per chart. Rows that miss the threshold stay visible and every chart has a “Show the data” table. An unknown tracked signal counts as its training average, not as “not true”.

−0.46 Of 16 everyday habits, only **going to bed 1 h+ later than usual** keeps a clear link with next-morning readiness (16 participants).

Mean absolute error, March 2020 hold-out (readiness, 0–10 scale)

Expected value (usual level + 13 tracked signals)	**1.01**
Usual level alone (last check-in in each of four earlier weeks)	**1.01**
Every check-in in those four weeks	**0.97**
A check-in every day (median of the last five)	**0.91**

The page reports two findings instead of claiming a good model: tracked data barely improves the expectation, and one check-in from each earlier week cost little accuracy against all of them — not yet a test of an app that asks once a week.

4 Limitations and next steps

Sixteen people, coarse self-report scales and associations that cannot show cause; the energy ledger is a concept demo with made-up parameters. A real version needs my own long-term data, retraining and validation, then a link to the cognitive core so an AI can join the review with clear sources, uncertainty and time ranges.

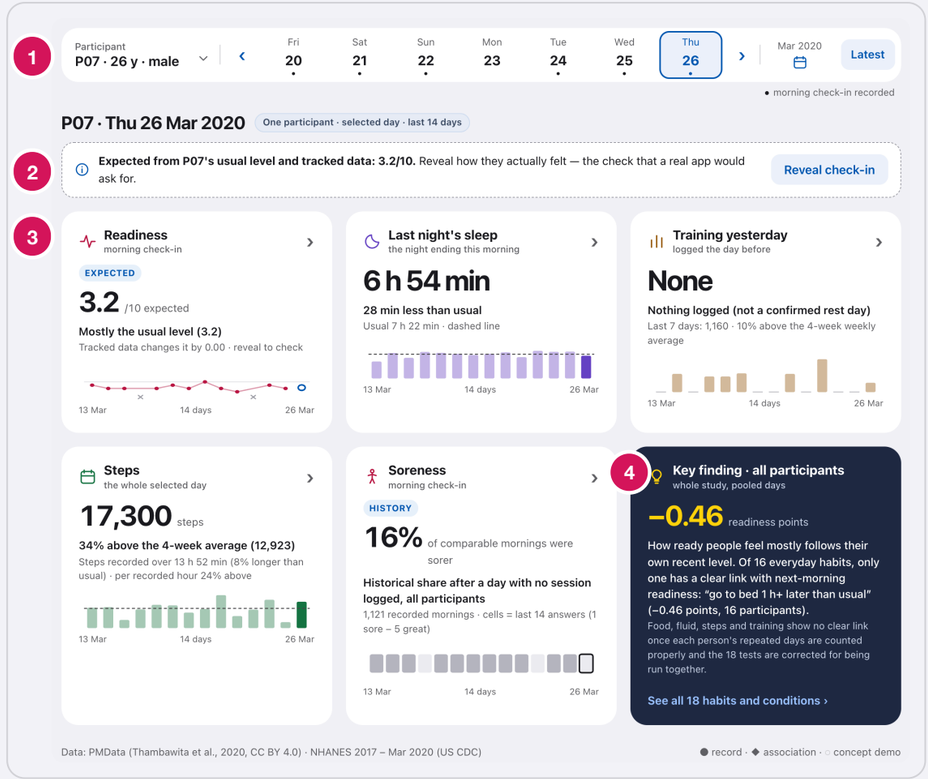


Fig. 1 · Today for P07 on Thu 26 Mar 2020: control card, the expectation to check, five summary cards (each naming its time range) with 14-day sparklines, and the key finding.

How to use it

1. Pick a participant, then a day on the 7-day strip; the calendar jumps to any date and **Latest** to the last recorded day.
2. Read the expected readiness, then press **Reveal check-in** to compare it with how the person felt.
3. Open any card for its detail view; the back button sits in the tab bar at the bottom. Hover a chart for exact values.
4. The key finding leads to **Everyday habits**; **Compare habits** sets the participant among the others.

5 Structure and interaction

The layout follows a phone app: the watch is the main data source, and only the phone can read Apple Health and reach my database. The floating frosted-glass tab bar follows Apple's current design language and holds the back button. **Today** follows “overview first, details on demand” (Shneiderman, 1996). Because one date mixes a morning check-in, last night, yesterday's training and a whole day of steps, every card names its time range. Readiness first shows an *expected* value — a usual level (the last check-in from each of the four weeks 8–35 days earlier) plus 13 tracked signals from the training log and Fitbit, built step by step in an animated waterfall — and **Reveal check-in** then shows what the person reported (Fig. 2). Five same-day questionnaire items were dropped because they are filled in with the readiness answer and would leak it.

Everyday habits compares 18 habits and states on one shared axis, grouped by when they happen or sorted by average difference (Fig. 3); **Compare habits** shows each person's numerators, denominators and position without ranking anyone. Date strip, calendar, picker and chart dates share one state; the training-load chart inspects a day before opening it (Fig. 4).

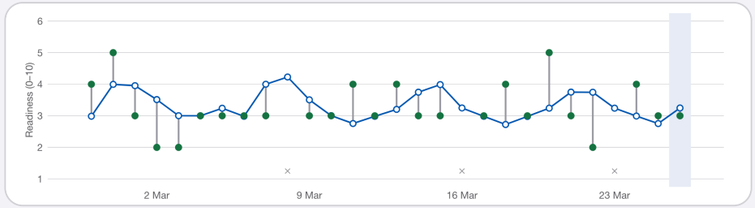


Fig. 2 · Felt vs expected, after Reveal. Blue: expected; green: the check-in; grey stick: the gap. The shaded band is the selected day.

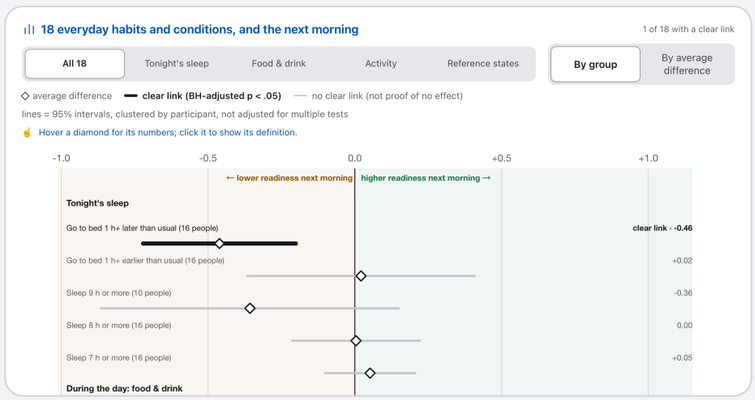


Fig. 3 · Everyday habits on one shared axis, by group or sorted by average difference (first rows shown). Thick black + “clear link” = BH-adjusted p < .05; thin grey = no clear link. Lines are unadjusted 95% intervals.

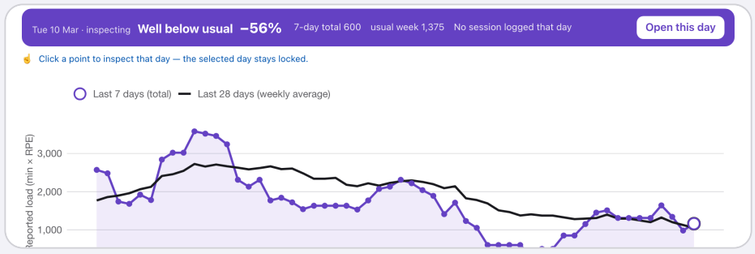


Fig. 4 · Inspect, then open. Clicking 10 Mar only inspects it (purple bar); the selected day stays locked until “Open this day”.

6 Visual choices

Visual hierarchy follows intellectual hierarchy: today's state before model detail, conclusions before full records. White cards and spacing group each question (Gestalt proximity), and scope chips name the unit of analysis. Blue marks the selected day or participant, green the real check-in and the better direction, amber the worse direction; colour is always paired with text or shape, so it is a pre-attentive cue, never the only encoding. Method notes are folded away to cut non-data ink (Tufte, 2001). In the forest plots the diamond answers “true on the selected day?” and the line weight answers “clear link?”. “No clear link” is labelled as not proof of no effect. The body figure shows a comparison group's history, not a forecast, for seven coarse questionnaire areas on a fixed 0–30%+ scale.

Shneiderman, B. (1996). The eyes have it. *IEEE Symposium on Visual Languages*. · Tufte, E. R. (2001). *The Visual Display of Quantitative Information* (2nd ed.). · Benjamini, Y. & Hochberg, Y. (1995). *JRSS B*, 57(1).

All data, software, images and AI assistance, and how each is used: Appendix, page 2.

Appendix · Resources and how they are used in the interface

Ready to Train? · GEOM90007 Assignment 2 · every resource in the submission, where it comes from, and where it appears

Jiakuo Tang

Resource	Source and licence	How it is used in the interface
Data		
PMData · Fitbit `sleep.json`	Thambawita et al. (2020), *PMData: a sports logging dataset*, ACM MMSys, doi:10.1145/3339825.3394926; Simula Research Laboratory; CC BY 4.0	Last night's sleep card and Sleep detail (sleep window, efficiency); sleep signals in the expected readiness; sleep rows of Everyday habits; sleep measures in Compare habits.
PMData · Fitbit `steps.json`, `resting_heart_rate.json`	As above	Steps card and hourly heatmap (with the time steps were recorded); step rows of Everyday habits; resting heart rate in the Sleep detail comparison and in Compare habits.
PMData · PMSys `wellness.csv`	As above	Morning check-ins: readiness (the revealed value and the usual level), soreness answers and soreness areas (body figure), and fatigue, mood, stress and sleep quality in the Sleep detail.
PMData · PMSys `srpe.csv`	As above	Training sessions (minutes × RPE load, activity tags): Training card and detail, rolling-load chart, monthly sessions, soreness after each load level, training signals and rows, energy-ledger demo.
PMData · `reporting.csv`, `participant-overview.xlsx`	As above	Daily food questionnaire (meals, alcohol, fluid) for the food rows of Everyday habits and Compare habits; age and sex for the participant picker, the body figure and NHANES matching.
NHANES 2017 – Mar 2020 · `P_DEMO`, `P_SLQ`	US CDC, National Center for Health Statistics; public domain	Compare habits → "Outside context": usual sleep reported by US adults of the same sex and similar age, set beside the participant's Fitbit nights as background only.
Derived files and model (in the submission)		
`daily.csv`, `sleep_episodes.csv`, `hourly_steps.csv`, `training_sessions.csv`, `questionnaires.csv`, `body_regions.csv`, `participants.csv`, `nhanes_sleep.csv`	Prepared from the raw PMData and NHANES archives by preparation scripts kept outside the submission (they need the raw archives to run); source hashes and rules are recorded in the included `*_provenance.json` files. No values are imputed.	Read once at start-up by `app.R`; every chart, card and table is computed from them. The raw archives are not needed to run the app.
`readiness_waterfall_model.rds`, `…_metrics.csv`, `…_metadata.json`	Trained by `train_readiness_waterfall.R` (ridge regression, λ = 3000; Nov–Feb fit, March hold-out). The app checks MD5 hashes of the inputs and flags a stale model.	The expected readiness, its waterfall, the felt-vs-expected chart and the accuracy line with its footnote.
Software		
R 4.6 · shiny · bslib	CRAN packages (open source)	Application framework, reactive state, Bootstrap 5 theme; bootstrap-datepicker (bundled with `shiny::dateInput`) for the calendar button.
ggplot2 · ggiraph · scales · systemfonts	CRAN packages (open source)	Every chart except the waterfall, drawn as interactive SVG: tooltips, click to open or inspect a day, click a diamond for its definition, click a participant to compare.
echarts4r	CRAN package; Apache ECharts (Apache 2.0)	The animated waterfall that builds the expected readiness step by step, with a replay button.
htmltools · htmlwidgets · jsonlite	CRAN packages (open source)	Custom HTML for cards, sparklines and tables; JSON for chart metadata and the model metadata.
`style.css`, `ui.js`	Written for this project	Layout, cards and frosted-glass tab bar; URL routing so the browser Back button works; focus management and chart names for screen readers.
Images, icons and fonts		
Female body figure `body-female-reference.png`	AI-generated for this project with ChatGPT (OpenAI image generation); approved by the author before use	Soreness → "After a day like yesterday, where were people most often sore?" for female participants.
Male body figure `body-male-reference.png`	AI-generated by the author with ChatGPT (OpenAI image generation), using a 3D body model as the reference; background converted to white	The same module for male participants.
Area masks `body-mask-*.png` **(14)**	Extracted from the two figures with an OpenCV thresholding script (kept outside the submission); recorded in `body_mask_provenance.json`	Shade the seven questionnaire areas on the body figures by the share of check-ins that ticked them.
Icons	Simple line icons drawn as inline SVG for this project	Card titles, tab bar and info links.
Fonts	The system font stack (San Francisco, Segoe UI, Helvetica Neue, Arial); no web fonts	All text and chart labels.
Design and method references		
Apple Health and Apple Fitness	Public interface patterns only; no Apple assets, icons or proprietary scores	Card summaries, the 7-day date strip, inspect-then-open on the training chart, the floating tab bar.
Shneiderman (1996); Tufte (2001); Gestalt and pre-attentive principles	Course readings	Overview first with details on demand; folded method notes; grouping by proximity; colour always paired with text or shape.
Benjamini & Hochberg (1995); cluster-robust (CR1) intervals; ridge regression	Standard statistical methods, implemented in base R	Everyday habits and "How did days like this usually go?" (clustered intervals, BH correction); the expected readiness (ridge).
AI assistance		
OpenAI Codex · Anthropic Claude	Coding assistants used under the author's direction; the header of `app.R` states which versions each assisted	Code, data preparation and analysis; they also updated this summary's wording and screenshots to match the final app. The author set the design goals, reviewed every result, and decided each iteration and all final design choices.